

Helix Thready — Icon Export Size/Format Matrix

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Rev	Date	Author	Change
1	2026-07-22	swarm (design · assets)	Initial matrix: exact per-platform pixel/format table for every OS surface (Web/PWA favicon + maskable, Android mipmap/adaptive/monochrome, iOS AppIcon incl. dark/tinted, Windows <code>.ico</code> , macOS <code>.icns</code> , Linux hicolor, HarmonyOS layered, Aurora); source-SVG→target mapping; output inventory tree; renderer/bundler tooling contract. Locked 1:1 to <code>generate-raster.sh</code> .
2	2026-07-22	swarm (design · assets · verify)	Verification pass: all 7 master SVGs confirmed <code>xmllint</code> -valid; <code>generate-raster.sh</code> confirmed <code>shellcheck</code> -clean and executed end-to-end (chromium renderer, ImageMagick <code>.ico</code> bundler, <code>.icns</code> correctly SKIPPED with note) — 83 real rasters, no faked/blank output (<code>favicon-16</code> non-blank, all outputs transparent-bg, <code>.ico</code> multi-res 16–256 verified). Corrected <code>logo-full-h64</code> cell 170→ 169 px ([64·1060/400]) to keep the 1:1 lock with the script. [VERIFIED — 2026-07-22 live run]

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1. Scope, sources of truth, and conventions

This file is the **exact pixel/format matrix** for every OS/platform icon Helix Thready ships, and is the specification `generate-raster.sh` (sibling in this directory) implements 1:1. It expands the platform tables in `brand-assets.md` §5 into concrete filenames and pixel dimensions.

Sources of truth (never contradicted, per **CONVENTIONS** §1):

- `brand-assets.md` — icon concept, numeric geometry (§3.1), platform matrix (§5), concrete manifests (§5.1). `[VERIFIED — inspected]`
- `design-system.md` §3.2 — the brand palette (`brand #B6E376` , `brand-2 #ABDDC9` / `#B7EBD6` , `accent #446E12` / `#B6E376`). `[VERIFIED]`
- `helix_thready_research_request_final.md` (decision matrix, Q1–Q45) and `helix_thready_subsystem_gaps_and_improvements.md` (gap register) — the merged request and the per-subsystem gaps referenced below (`[GAP: 8.5]`). `[CONSTITUTION]`

Units. All sizes are **pixels of the exported raster** unless the row says “vector”. Android adaptive/monochrome are quoted as **dp @ density** with the resolved px in the same cell (px = `dp × density` , mdpi = ×1). iOS legacy rows quote the resolved **px** (the `pt@scale` role is noted).

Provenance legend (per **CONVENTIONS** §3): `[VERIFIED]` size set is the platform-canonical set corroborated by `brand-assets.md` §5 / platform packaging docs · `[RESEARCH]` must be re-verified against current OS docs at integration · `[GAP: id]` produced now but consumed by a subsystem the gap register flags as a **scaffold** (do not claim that launcher ships).

2. Source SVGs and the source → target mapping

Six hand-authored, **valid** (xmllint-clean) master vectors in this directory feed every raster. No output invents a new drawing — each is a rasterization of one master (single-source consistency, `brand-assets.md` §3).

Source SVG	Fill	Transparent bg	Feeds
<code>launcher-icon.svg</code> (neutral master)	brand #B6E376 → brand-2 #ABDDC9 gradient	yes	Web/PWA, Android legacy+adaptive- fg+Play, iOS light + legacy set, Windows, macOS, Linux, HarmonyOS, Aurora
<code>launcher-icon-light.svg</code>	accent #446E12 → brand #B6E376	yes	light-surface in- app mark; optional light launcher slot
<code>launcher-icon-dark.svg</code>	brand #B6E376 → brand-2(dar k) #B7EBD6	yes	iOS 18 dark appearance (<code>icon-1024- dark.png</code>); dark- surface in-app mark
<code>launcher-icon-mono.svg</code>	single tone via currentColor (default accent #446E12)	yes	Android monochrome layer; iOS 18 tinted appearance (<code>icon-1024- tinted.png</code>)
<code>logo-full.svg</code> (mark + “Thready” wordmark)	mark gradient + theme- aware text	yes	headers/ marketing lockup raster (§3.9) — never a launcher icon
<code>logo-mark.svg</code> (tight-crop mark)	brand → brand-2 gradient	yes	avatar/lockup raster (§3.9)
<code>footer-slogan.svg</code> (“Made with ♥ by Helix Development”)	heart ds- heart = accent ; text muted	yes	footer lockup raster (§3.9)

The launcher icons are **letter-free by construction** (brand-assets.md §1); the wordmark and slogan carry text and are never baked into a launcher/OS icon slot.

3. The pixel/format matrix (per platform)

3.1 Web / PWA

Output file	Size (px)	Format	Source	Purpose	Provenance
<code>web/favicon.svg</code>	vector	SVG	<code>launcher-icon.svg</code>	scalable favicon (<code>rel=icon type=image/ svg+xml</code>)	[VERIFIED]
<code>web/favicon-16.png</code>	16	PNG	master	favicon component	[VERIFIED]
<code>web/favicon-32.png</code>	32	PNG	master	favicon component	[VERIFIED]
<code>web/favicon-48.png</code>	48	PNG	master	favicon component	[VERIFIED]
<code>web/favicon.ico</code>	16 + 32 + 48	ICO (multi)	master (bundled)	legacy <code>rel=icon</code>	[VERIFIED]
<code>web/icon-192.png</code>	192	PNG	master	PWA manifest <code>purpose:any</code>	[VERIFIED]
<code>web/icon-512.png</code>	512	PNG	master	PWA manifest <code>purpose:any</code>	[VERIFIED]
<code>web/ maskable-192.png</code>	192	PNG	master	PWA <code>purpose:maskable</code> (art within Ø916 safe zone, §3.1)	[VERIFIED]
<code>web/ maskable-512.png</code>	512	PNG	master	PWA <code>purpose:maskable</code>	[VERIFIED]
<code>web/apple-touch- icon.png</code>	180	PNG	master	iOS Safari home-screen	[VERIFIED]

Wired via `manifest.webmanifest` + `<head>` in `brand-assets.md` §5.1 (self-hosted, no external CDN — CSP hygiene).

3.2 Android

Legacy density mipmaps (`res/mipmap-<density>/ic_launcher.png`):

Density	Size (px)	Output	Source	Provenance
mdpi (×1)	48	<code>android/ mipmap-mdpi/ ic_launcher.png</code>	master	[VERIFIED]
hdpi (×1.5)	72	<code>android/ mipmap-hdpi/ ic_launcher.png</code>	master	[VERIFIED]
xhdpi (×2)	96	<code>android/ mipmap-xhdpi/ ic_launcher.png</code>	master	[VERIFIED]
xxhdpi (×3)	144	<code>android/ mipmap-xxhdpi/ ic_launcher.png</code>	master	[VERIFIED]
xxxhdpi (×4)	192	<code>android/ mipmap- xxxhdpi/ ic_launcher.png</code>	master	[VERIFIED]

Adaptive **foreground** + Android-13 **monochrome** (108 dp canvas, `res/mipmap-<density>/`):

Density	108 dp → px	<code>_foreground</code> source	<code>_monochrome</code> source	Provenance
mdpi	108	master	<code>launcher-icon-mono.svg</code>	[VERIFIED]
hdpi	162	master	mono	[VERIFIED]
xhdpi	216	master	mono	[VERIFIED]
xxhdpi	324	master	mono	[VERIFIED]
xxxhdpi	432	master	mono	[VERIFIED]

Extra	Size (px)	Output	Notes
Adaptive background	—	<code>@color/ ic_launcher_background</code>	solid/token color, not a raster (brand-assets.md §5.1 <code>ic_launcher.xml</code>)
Play Store listing	512	<code>android/play-store-512.png</code>	[VERIFIED]

Adaptive art stays within the **66 dp inner safe circle** so no coil clips under the OS mask (§3.1).

`ic_launcher.xml` (`mipmap-anydpi-v26`) references `background` / `foreground` / `monochrome` verbatim per `brand-assets.md` §5.1.

3.3 iOS

`Assets.xcassets/AppIcon.appiconset/` — modern single-size + iOS 18 appearances:

Output file	Size (px)	Source	Appearance	Provenance
<code>icon-1024.png</code>	1024	<code>launcher-icon.svg</code>	any / light	[VERIFIED]
<code>icon-1024-dark.png</code>	1024	<code>launcher-icon-dark.svg</code>	dark (<code>luminosity:dark</code>)	[VERIFIED]
<code>icon-1024-tinted.png</code>	1024	<code>launcher-icon-mono.svg</code>	tinted (<code>luminosity:tinted</code>)	[VERIFIED]

Legacy full set (only if targeting pre-Xcode-14 catalogs), all from master:

Size (px)	<code>pt@scale</code> role
20	notification 20pt@1x (iPad)
29	settings 29pt@1x
40	notification 20pt@2x / spotlight 40pt@1x
58	settings 29pt@2x
60	notification 20pt@3x
76	app 76pt@1x (iPad)
80	spotlight 40pt@2x
87	settings 29pt@3x
120	app 60pt@2x (iPhone) / spotlight 40pt@3x
152	app 76pt@2x (iPad)
167	app 83.5pt@2x (iPad Pro)
180	app 60pt@3x (iPhone)

`Contents.json` (single-size + dark + tinted) is reproduced verbatim in `brand-assets.md` §5.1.

[VERIFIED]

3.4 Windows

`.ico` multi-resolution bundle (Tauri `icon` / `tauri.conf.json`):

PNG components (px)	Bundle	Source	Provenance
16, 24, 32, 48, 64, 128, 256	<code>windows/ thready.ico</code>	master	[VERIFIED]

3.5 macOS

`.icns` iconset components (`macos/thready.iconset/`), bundled to `macos/thready.icns`:

Iconset name	Size (px)	Source
<code>icon_16x16.png</code>	16	master
<code>icon_16x16@2x.png</code>	32	master
<code>icon_32x32.png</code>	32	master
<code>icon_32x32@2x.png</code>	64	master
<code>icon_128x128.png</code>	128	master
<code>icon_128x128@2x.png</code>	256	master
<code>icon_256x256.png</code>	256	master
<code>icon_256x256@2x.png</code>	512	master
<code>icon_512x512.png</code>	512	master
<code>icon_512x512@2x.png</code>	1024	master

Unique pixel sizes bundled: **16, 32, 64, 128, 256, 512, 1024.** [VERIFIED]

3.6 Linux

hicolor icon theme (`linux/hicolor/<size>x<size>/apps/thready.png`) + scalable SVG:

Size (px)	Output	Source	Provenance
16	<code>hicolor/16x16/apps/thready.png</code>	master	[VERIFIED]
22	<code>hicolor/22x22/apps/thready.png</code>	master	[VERIFIED]
24	<code>hicolor/24x24/apps/thready.png</code>	master	[VERIFIED]
32	<code>hicolor/32x32/apps/thready.png</code>	master	[VERIFIED]
48	<code>hicolor/48x48/apps/thready.png</code>	master	[VERIFIED]
64	<code>hicolor/64x64/apps/thready.png</code>	master	[VERIFIED]
128	<code>hicolor/128x128/apps/thready.png</code>	master	[VERIFIED]
256	<code>hicolor/256x256/apps/thready.png</code>	master	[VERIFIED]
512	<code>hicolor/512x512/apps/thready.png</code>	master	[VERIFIED]
vector	<code>hicolor/scalable/apps/thready.svg</code>	master (copy)	[VERIFIED]

3.7 HarmonyOS

Layered image (`resources/base/media/`), consumed by the ArkTS client — a **scaffold** `[GAP: 8.5 helix_shims / HarmonyOS]` ; assets are produced now, but no HarmonyOS launcher is claimed to ship:

Output	Size (px)	Source	Role	Provenance
<code>harmonyos/foreground.png</code>	216	master	<code>layered_image.json > foreground</code>	[GAP: 8.5]
<code>harmonyos/background.png</code>	216	master	<code>layered_image.json > background</code> (solid/branded bg supplied at integration)	[GAP: 8.5]
<code>harmonyos/appgallery-1024.png</code>	1024	master	AppGallery listing	[GAP: 8.5]

`layered_image.json` is reproduced in `brand-assets.md` §5.1.

3.8 Aurora

Aurora OS (auroraos.ru / Sailfish density buckets) — [RESEARCH], re-verify against current Aurora packaging docs at integration ([OPEN: THREADY-DES-05]); consumed by a native Qt client scaffold [GAP: 8.5]:

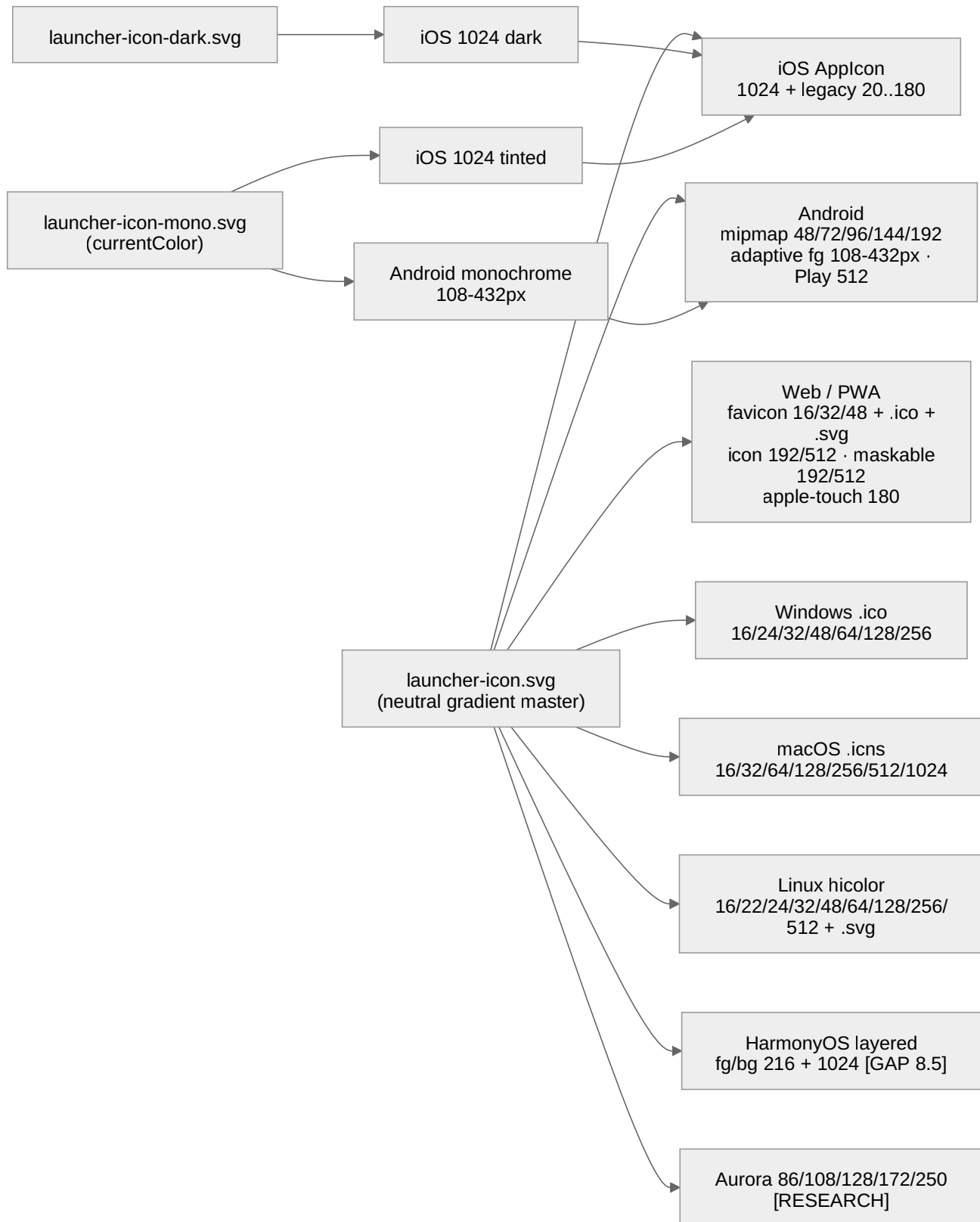
Output	Size (px)	Source	Provenance
<code>aurora/thready-86.png</code>	86	master	[RESEARCH]
<code>aurora/thready-108.png</code>	108	master	[RESEARCH]
<code>aurora/thready-128.png</code>	128	master	[RESEARCH]
<code>aurora/thready-172.png</code>	172	master	[RESEARCH]
<code>aurora/thready-250.png</code>	250	master	[RESEARCH]

3.9 Brand lockups (non-launcher, convenience raster)

Emitted for docs/marketing/email where an SVG cannot be used. **Not** launcher icons (they carry text); never placed in an OS icon slot.

Output	Size (W×H px)	Source	Notes
<code>brand/logo-full-h64.png</code>	169×64	<code>logo-full.svg</code>	aspect 1060:400 preserved (169 = $\lfloor 64 \cdot 1060 / 400 \rfloor$)
<code>brand/logo-full-h128.png</code>	339×128	<code>logo-full.svg</code>	aspect preserved
<code>brand/logo-mark-128.png</code>	128×128	<code>logo-mark.svg</code>	square mark
<code>brand/logo-mark-256.png</code>	256×256	<code>logo-mark.svg</code>	square mark
<code>brand/logo-mark-512.png</code>	512×512	<code>logo-mark.svg</code>	square mark
<code>brand/footer-slogan-h44.png</code>	560×44	<code>footer-slogan.svg</code>	aspect 560:44 preserved

4. Export fan-out (diagram)



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `icon-export-fanout.mmd`.

Explanation (for readers/models that cannot see the diagram). Three of the six master vectors drive the launcher outputs. The **neutral master** (`launcher-icon.svg`) is the single source for almost every platform bundle: the Web/PWA set (scalable + 16/32/48 favicons, the multi-resolution `.ico` , the 192/512 `purpose:any` icons, the 192/512 `maskable` icons whose art sits inside the Ø916 safe zone, and the 180 px apple-touch icon); the Android legacy density mipmaps (48/72/96/144/192) and the adaptive **foreground** across the five densities (108→432 px) plus the 512 Play-Store image; the iOS 1024 base and its legacy 20–180 px set; the Windows `.ico` components (16–256); the macOS `.icns` iconset (unique sizes 16–1024); the Linux hicolor theme (16–512 plus the scalable SVG copy); and the HarmonyOS layered foreground/background (216) + AppGallery (1024) and Aurora buckets (86–250). Two masters specialize appearance-variant slots: the **dark master** (`launcher-icon-dark.svg`) renders the iOS 18 **dark** 1024, and the **monochrome master** (`launcher-icon-mono.svg` , drawn with `currentColor`) renders both the Android 13+ **monochrome** layer at every density and the iOS 18 **tinted** 1024. The monochrome and dark/tinted branches feed back into their platform’s asset set (they are additional slots in the same Android `mipmap` folders and the same iOS `AppIcon.appiconset`), which is why their edges rejoin the Android and iOS nodes. Because every arrow originates from one hand-authored, xmlint-valid vector, a change to a master re-emits its whole downstream fan without drift. HarmonyOS and Aurora are drawn but flagged (`[GAP 8.5]` , `[RESEARCH]`) because the clients that consume them are scaffolds.

5. Output inventory tree

`generate-raster.sh [OUTPUT_DIR]` (default `./raster`) writes exactly this tree:

```
raster/
  web/      favicon.svg favicon-{16,32,48}.png favicon.ico
            icon-{192,512}.png maskable-{192,512}.png apple-touch-icon.png
  android/  mipmap-{mdpi,hdpi,xhdpi,xxhdpi,xxxhdpi}/ic_launcher.png
            mipmap-{mdpi,hdpi,xhdpi,xxhdpi,xxxhdpi}/ic_launcher_foreground.png
            mipmap-{mdpi,hdpi,xhdpi,xxhdpi,xxxhdpi}/ic_launcher_monochrome.png
            play-store-512.png
  ios/AppIcon.appiconset/
            icon-1024.png icon-1024-dark.png icon-1024-tinted.png
            icon-{20,29,40,58,60,76,80,87,120,152,167,180}.png
  windows/  icon-{16,24,32,48,64,128,256}.png thready.ico
  macos/thready.iconset/
            icon_16x16.png ... icon_512x512@2x.png          (10 components)
            thready.icns                                     (if packer present)
  linux/hicolor/
            {16,22,24,32,48,64,128,256,512}x.../apps/thready.png
            scalable/apps/thready.svg
  harmonyos/ foreground.png background.png appgallery-1024.png [GAP 8.5]
  aurora/    thready-{86,108,128,172,250}.png [RESEARCH]
  brand/     logo-full-h{64,128}.png logo-mark-{128,256,512}.png footer-slogan-h44.png
```

Total distinct raster targets ≈ **95 PNGs + 2 SVG copies + up to 2 bundles** (`.ico` , `.icns`).

6. Renderer & bundler tooling contract

`generate-raster.sh` **detects** a real rasterizer and **self-tests** it before use; if none renders it prints a `[SKIP]` note and exits 0 **without writing any placeholder/fake PNG** (the “no bluff” rule, [CONVENTIONS §7](#)).

Role	Accepted tools (in preference order)	If absent
SVG → PNG rasterizer	<code>rsvg-convert</code> (librsvg) → <code>inkscape</code> → <code>resvg</code> → headless Chromium/Chrome → ImageMagick*	<code>[SKIP]</code> , exit 0, no output faked
<code>.ico</code> bundler	<code>icotool</code> → ImageMagick (<code>magick</code>)	PNG components kept; <code>.ico</code> step skipped with a note
<code>.icns</code> bundler	<code>png2icns</code> → <code>iconutil</code> (macOS)	<code>.iconset/*.png</code> kept; run <code>iconutil -c icns</code> on macOS

* ImageMagick is accepted **only** if its SVG delegate actually renders this directory’s masters: the probe rejects an IM whose internal MSVG coder mishandles `<text> /CSS/ currentColor`, so it falls through to the next renderer instead of emitting garbage. [\[VERIFIED – probe logic in the script\]](#)

Detected in this environment [\[VERIFIED – 2026-07-22\]](#): `magick` / `convert` and headless `chromium` are present; `rsvg-convert`, `inkscape`, `resvg`, `icotool`, `png2icns`, `iconutil` are **absent**. The script therefore selects Chromium (which renders the masters and honors the transparent background via `--default-background-color=00000000`); `.ico` bundling falls to ImageMagick; `.icns` bundling is **skipped with a note** (no `png2icns` / `iconutil` here) and leaves the `.iconset` PNGs for a macOS packaging pass. All seven source SVGs are xmllint-valid, so the strict librsvg/resvg paths also succeed where those tools are installed.

7. Verification / legibility gates

Per `brand-assets.md §6` export-pipeline `verify:` block and `design-system.md §8`:

- **Legibility:** every rendered size asserts no clipped coil (art within the Ø768 keyline / Ø916 maskable safe circle) and $\geq 3:1$ mark/background where placed. The **full** gradient tier is used ≥ 48 px; 16–32 px favicons rely on the simplified/heavier-stroke reading (mono stroke widened to 128 for tint slots).
- **Transparent background:** every launcher output is transparent (probe render confirms non-opaque canvas); no platform corners are baked into the art — the OS applies its own mask.
- **Visual-regression:** snapshot each target into the `ScreenDiff` bank [\[GAP: 9.3\]](#) so a master edit that breaks a size fails CI once that bank has CI (workable item `THREADY-DES-VR-01`).

8. Gaps & open items

- `[GAP: 8.5 helix_shims / HarmonyOS + Aurora]` — HarmonyOS layered + Aurora PNGs are produced now but consumed by native ArkTS/Qt clients that are **scaffolds**; do not claim those launchers ship.
 - `[OPEN: THREADY-DES-05]` — Aurora density buckets (86/108/128/172/250) are `[RESEARCH]`; re-verify against current Aurora OS packaging docs at integration.
 - `[OPEN: THREADY-DES-03]` — heart color in `footer-slogan.svg` is `ds-heart = accent` (brand) by in-house precedent; classic love-red is the alternative pending an operator decision.
 - `[GAP: 9.3 VisualRegression family]` — the size-legibility snapshots have no CI yet; tracked as THREADY-DES-VR-01.
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